

4.2.4. Most Frequently Sold Product Pairs

15:26

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explanation:

Transactions:

- **2025-01-01:** Product A, Product B
- **2025-01-02:** Product A, Product C
- **2025-01-03:** Product B, Product A
- **2025-01-04:** Product C, Product B
- **2025-01-05:** Product A, Product C

Now, let's count how often the pairs of products appear together:

- **Product A and Product B:** Appear in transactions on 2025-01-01 and 2025-01-03.
- **Product A and Product C:** Appear in transactions on 2025-01-02 and 2025-01-05.
- **Product B and Product C:** Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- **Product A and Product B** (2 times)
- **Product A and Product C** (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Code:

```
import pandas as pd

from itertools import combinations

from collections import Counter


# Prompt user to input the file name

file_name = input()


# Read data from the specified CSV file

df = pd.read_csv(file_name)


# write the code

grouped = df.groupby("Date")["Product"].apply(list)

pairs_counter = Counter()

for products in grouped:

    pairs = combinations(sorted(products), 2)

    #Sort to ensure consistency

    pairs_counter.update(pairs)

#Find the most common product pairs

max_frequency = max(pairs_counter.values())

most_common_pairs = [(pair, freq)

    for pair, freq in pairs_counter.items() if freq == max_frequency]

# Output the most frequent product pairs

for (product1, product2), frequency in most_common_pairs:

    print(f'{product1} and {product2}: {frequency} times')
```

Average time

0.085 s
85.00 ms

Maximum time

0.173 s
173.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1

173 ms

Debug

^

Expected output	Actual output
<code>sales_data.csv</code>	<code>sales_data.csv</code>
Product A and Product B: 2 times	Product A and Product B: 2 times
Product A and Product C: 2 times	Product A and Product C: 2 times